FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*

   Full Marks : ***+4*** *If only (all) the four option(s) is (are) chosen.*
   Partial Marks : ***+3*** *If all the four options are correct but ONLY three options are chosen.*
   Partial Marks : ***+2*** *If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   Partial Marks : ***+1*** *If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   Zero Marks : ***0*** *If none of the options is chosen (i.e. the question is unanswered).*

   Negative Marks : ***–1*** *In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

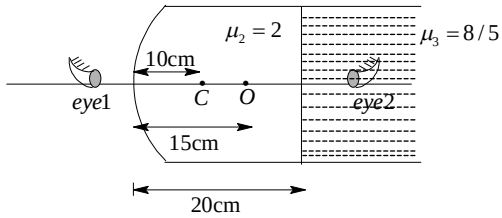
Physics

PART – I

SECTION – A
(One Options Correct Type)

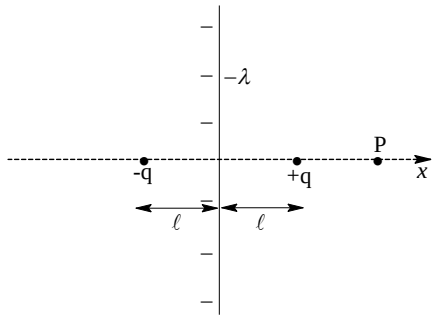
This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The separation between the images seen by the two eyes of the object O is (C is centre of curvature):



(A) 4cm
(B) 10cm
(C) 11cm
(D) 14cm

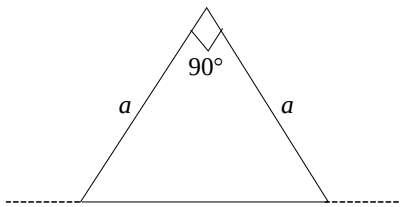
2. An infinite line charge is placed along y-axis and two point charges are placed on the x-axis as shown in the figure. Find x for which field on the positive x-axis is zero $\left(x >> \ell\right)$ and $\frac{q\ell}{\lambda} = 2$:



(A) 2 (B) 1
(C) $\frac{1}{2}$ (D) 4

3. A wooden block is partially submerged in a liquid contained in a vessel. Now the vessel is suddenly accelerated upwards with constant acceleration. Then:
(A) Volume of block submerged will start to increase
(B) Volume of block submerged will start to decrease
(C) Buoyant force will increase
(D) Buoyant force will decrease

4. The moment of inertia of an isosceles triangular plate of mass m and side a about an axis through the side shown in the figure is:



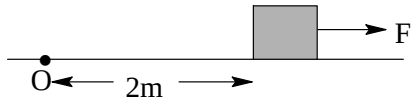
(A) $\frac{ma^2}{6}$ (B) $\frac{2ma^2}{3}$
(C) $\frac{ma^2}{4}$ (D) $\frac{ma^2}{12}$

Space for Rough work

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. A cube of side 1m and mass 1kg is pulled by a force F acting along the central line so that it slides with an acceleration of 1 m/s^2 on a rough surface $(\mu = 0.1)$.



Consider a point **O** on the rough surface. Then

(A) Torque of force F about O is 1N-m

(B) Torque of force F about O is $\frac{1}{2}\text{N-m}$

(C) Torque of all forces acting on the cube about O is $\frac{1}{2}\text{N-m}$

(D) Torque of all forces acting on the cube about O is 1N-m

6. In case of walking on a rough surface:

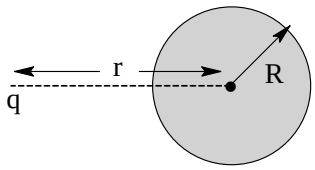
(A) Friction force acts in the direction of motion

(B) Friction force acts opposite to the direction of motion

(C) Change in kinetic energy is due to work done by friction

(D) Change in kinetic energy is due to work done by internal forces.

7. A point q is placed outside an uncharged conducting sphere as shown in the figure. Then:



(A) The potential of the sphere is zero.

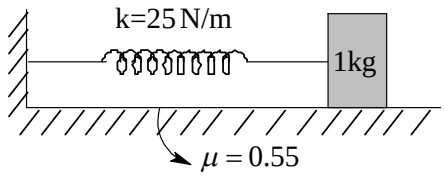
(B) The potential of the sphere is $\frac{kq}{r}$

(C) If the sphere is earthed charge appearing on its surface is $\frac{R}{r}(-q)$

(D) If the sphere is earthed charge appearing on its surface is $\frac{r}{R}(-q)$

Space for Rough work

8. One end of a spring of force constant 25 N/m is connected with a block of mass 1kg and the other end is fixed to a vertical wall. The block is given velocity 6 m/s towards the wall (when the spring is in natural length) on a horizontal rough surface having coefficient of friction 0.55. Then:



(A) The compression in the spring when the block stops momentarily for the Ist time is 1m.
(B) The compression in the spring when the block stops momentarily for the Ist time is 0.5m.
(C) The acceleration of the block when it stops momentarily for the IInd time is 12.5m/s^2
(D) The acceleration of the block when it stops momentarily for the IInd time is 8.5m/s^2

9. If the face of a woman is 8.5 inches wide, distance between pupil of her two eyes is 2.5 inches and top of head to chin seperation is 9 inches. Then the minimum dimension of a rectangular mirror so that she can see her complete face is:

(A) Height $= 4.5 \text{inches}$ (B) Height $= 2.25 \text{inches}$

(C) Width $= 3 \text{inches}$ (D) Width $= 3.5 \text{inches}$

10. A two particle system of masses 2 kg and 3 kg present at (15, 0) and (0, 20) in a x, y plane then its centre of mass lies on lines:

(A) $y = 2x$ (B) $\frac{x}{12} + \frac{y}{24} = 1$

(C) $3x - y = 6$ (D) $y = x$

11. If a cylinder performs rolling without slipping on a horizontal accelerated rough plank. Then:
(A) The mechanical energy of cylinder is conserved
(B) The mechanical energy of cylinder is not conserved.
(C) The mechanical energy of the system of plank and cylinder is conserved.
(D) The mechanical energy of the system of plank and cylinder is not conserved.

12. Two particles are projected from a horizontal plane with the same initial velocity v_0 at two different angles of projection θ_1 and θ_2, such that their ranges are the same. The ratio of their maximum heights reached is:

(A) $\tan^2\theta_1$ (B) $\theta = 60°$

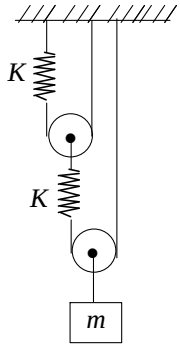
(C) $t = \frac{20}{\sqrt{3}} s$ (D) $t = \frac{10}{\sqrt{3}} s$

Space for Rough work

SECTION – C
(Numerical Answer Type)

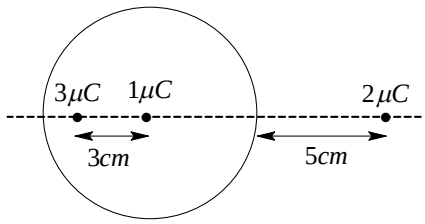
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A block of mass hangs in equilibrium with the help of two spring as shown in the figure. Find the time period of vertical oscillation of the block. Take m = 10kg and $K = 50N/m$: (in second)



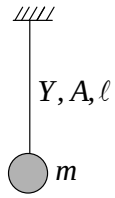
14. In an experiment for calculation of focal length of convex lens the distance between a screen and an object is 120 cm. A convex lens is placed close to the object and is moved along the line joining object and screen, towards the screen. If the ratio of magnification of two real images is $1:16$ then find the focal length of the convex lens: (in cm)

15. A spherical conducting shell of radius 5 cm is having a point charges placed inside and outside the shell as shown in figure. Find the electric force on $1\mu C$ at the centre due to induce charges on the outer surface of shell (in Newton)

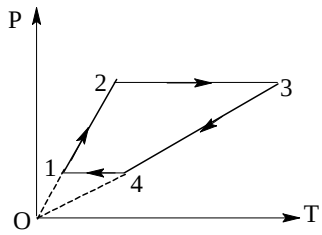


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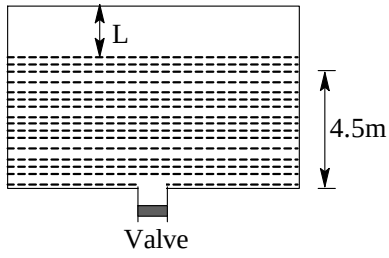
16. A spherical ball is hanging from an elastic rubber wire of young modulus Y, length ℓ and area A as shown in the figure. Find the time period of small vertical oscillations of the bob.
[Take $Y=10^6 N/m^2$, $A=1mm^2$, $\ell=1m$ and $m=1kg$]



17. Three moles of an ideal monoatomic gas performs a cyclic process as shown in the figure. The temperature in different states are $T_1=400k$; $T_2=800k$, $T_3=2400k$ and $T_4=1200k$. Determine the work done by the gas during the cycle (in KJ) [Given $R=8.31J/mole-KJ$]



18. The vertical water tank, shown has uniform cross section, closed at the top and initial level of water in it is 4.5 m from bottom. The empty space of length L contains air at atmospheric pressure (10^5 Pa), that can be considered as an ideal gas. When the value at the bottom is opened, the level of water decreases by 0.5 m when the flow of water ceases through valve remains opened. Neglecting any variation in temperature during the process, find initial length of empty space L in cm. (g = 10 m/s^2, density of water = 1000 kg/m^3)



Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Which of the following ion is most stable in aq. solution?

(A) NH_4^+ (B) H_2F^+

(C) H_2O^+ (D) CH_5^+

20. What are the possible oxidation state(s) of the Cr in the complex $K_2\left[Cr(NO)(NH_3)(CN)_4\right]$.

Given (magnetic moment) $\mu = 1.73\ BM$.

(A) +2 (B) +3

(C) +1 (D) All are possible

21. Which of the following statements concerning a weak acid-strong base titration is true?

(A) The pH after the equivalence point of the weak acid strong base titration is determined by using the K_b expression for the conjugate base

(B) A buffer solution of the weak acid and its conjugate base is formed before the equivalence point is reached

(C) The pH at the equivalence point of a weak monoprotic acid strong base titration is equal to the pH at the equivalence point of a strong acid-strong base titration

(D) The increase in pH in the region near the equivalence point of the weak acid strong base titration is greater than the pH change in the same region of a strong acid strong base titration

22. A hypothetical reaction

$A_2 + B_2 \longrightarrow 2AB$ Follows mechanism as given below:

$A_2 \underset{k_b}{\overset{k_c}{\rightleftharpoons}} A + A.....(fast)$

$A + B_2 \xrightarrow{k_1} AB + B.....(slow)$

$A + B \xrightarrow{k_2} AB.....(fast)$

The order of overall reaction is

(A) 2.5 (B) 1

(C) 3/2 (D) zero

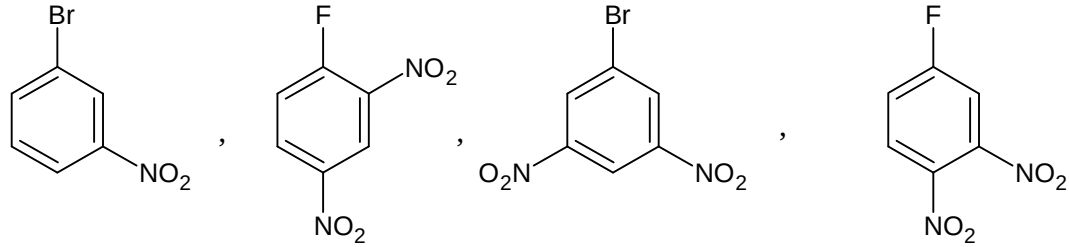
Space for Rough work

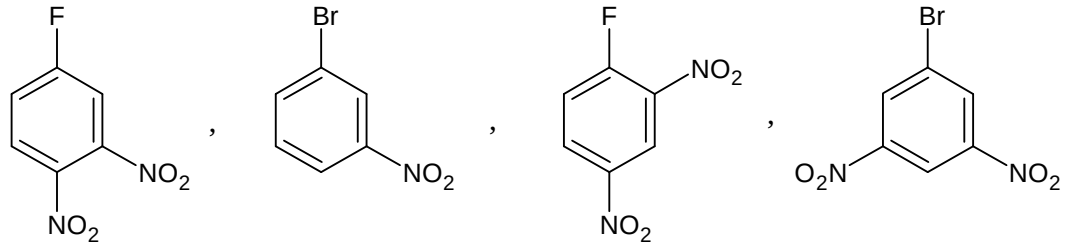
(One or More than one correct type)

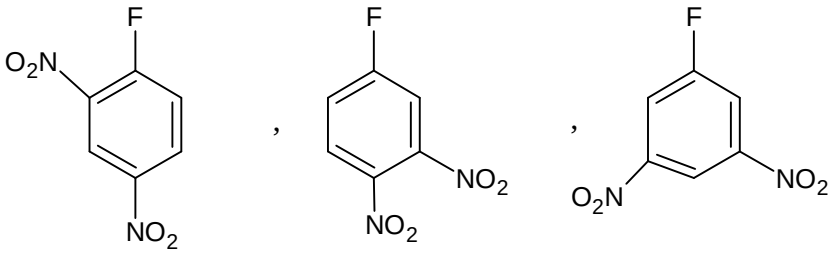
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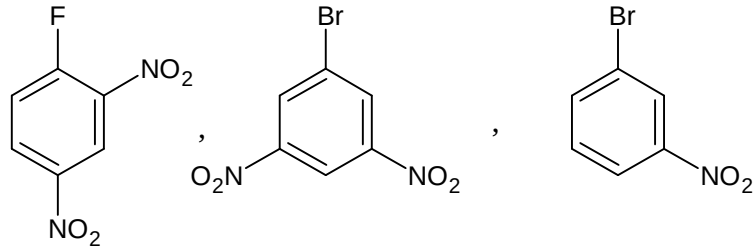
23. Which of the following statement is/are true?
(A) Alkali metals are soft metals and softness increases down the group
(B) Lattice energy of alkali metal salts with same anion decreases with increase in size of cations
(C) Paramagnetic character of $Na-Liq\ NH_3$ solution decreases in increase in the conc. of NH_3
(D) LiF has lowest solubility among all alkali metal halides

24. Which of the following correctly ranks the aryl halides in decreasing order of reactivity toward sodium methoxide in methanol?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

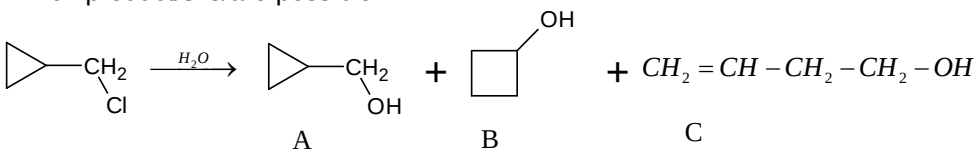
25. The solubility of alkaline earth metal salts in water decreases down the group, the salts are
(A) phosphates (B) sulphates
(C) carbonates (D) hydroxides

26. In the extraction of Li, a mixture of dry lithium chloride and potassium chloride is fused and electrolysed, the following fact(s) is/are correct regarding this
(A) KCl increases the conductivity
(B) M.P. of LiCl lowers due to presence of KCl
(C) Reduction potential of Li^+ is higher than K^+ in fused state
(D) Reduction potential of Li^+ is lower than K^+ in fused state

27. Which of the following is/are not a characteristic of effusion/diffusion?
(A) The rate of effusion/diffusion is inversely proportional to average speed of the gas particles
(B) Gases mix as a result of the random motion and frequent collision of gas particles
(C) The rate of effusion/diffusion decreases with increasing temperature
(D) Rate of effusion/diffusion is directly proportional to molar mass of the gas

28. Which of the following is/are characteristic of a tetrahedral complex?
(A) the $d_{x^2-y^2}$ and d_{z^2} orbitals are the low energy orbitals
(B) most tetrahedral complex are high spin
(C) crystal filed splitting is about twice found in octahedral complexes
(D) the splitting pattern is just opposite of that in octahedral complexes

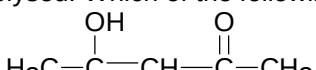
29. Which product/s is/are possible?

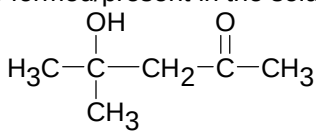


A B C

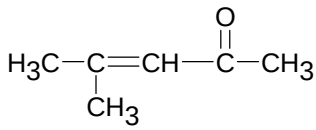
(A) A (B) B
(C) C (D) none of these

30. When one equivalent of CH_3MgBr is treated with $CH_3COCH_2COCH_3$ and subsequently acid hydrolysed. Which of the following compounds may be formed/present in the solution/container?

(A) 

(B) 

(C) CH_4

(D) 

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. One mole of an ideal gas is expanded isothermally at 298 K until its volume is tripled. Find the value of ΔS_{sys} (entropy of system) if the expansion is a free expansion (in vacuum). Given (log3= 0.477 R = 8.3 $JK^{-1}mol^{-1}$). Give your answer in JK^{-1} upto two decimals only.

32. Gold metal is extracted from its ore by treating the crushed rock with an aerated cyanide solution. The unbalanced equation for the reaction is

$$Au(s) + CN^{-}(aq) + O_2(g) \rightarrow Au(CN)_2^{-}(aq)$$

Use any of the following data at 27^oC to calculate ΔG° for this reaction in basic solution.

Given K_f for $Au(CN)_2^- = 10^{38}$

$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(\ell)$ $\quad E^\circ = 1.2$ V

$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$ $\quad E^\circ = 1.5$ V

$Au^{3+}(aq) + 2e \rightarrow Au^+$ $\quad E^\circ = 1.5$ V

Use equation $\Delta G^\circ = -2RT\log_{10} Q$ and R = $8\,Jmol^{-1}K^{-1}$, F = 96500 C. Give your answer (magnitude only) in kJ.

33. At certain temperature the dissociation constants of formic acid and acetic acid are 1.8×10^{-4} and 1.8×10^{-5} respectively. The concentration of acetic acid solution(molarity) in which the hydrogen ion has the same concentration as 0.001 M formic acid solution is equal to

Space for Rough work

34. If 250 ml O_2 over water at 30ºC and a total pressure of 740 torr is mixed with 300 ml of N_2 over water at 25^oC at total pressure of 780 torr. What will be total pressure if the mixture is in a 500 ml vessel over water at 35ºC. (Given aq. tension of water at 25^oC, 30^oC and 35^oC are 23.8, 31.8 and 42.2 torr.)

35. The solubility product of $Co_2\left[Fe(CN)_6\right]$ in water is $K\times10^{-16}$. The value of K is

Given $\lambda^\circ_m Co^{+2} = 80\Omega^{-1}$ cm^{-1} mol^{-1}

$\lambda^\circ_m [Fe(CN)_6]^{-4} = 40\Omega^{-1}$ cm^{-1} mol^{-1}

Specific conductivity of saturated solution $= 2\times10^{-6}\Omega^{-1}c\ m^{-1}$

Specific conductivity of water used = $= 4\times10^{-7}\Omega^{-1}c\ m^{-1}$

36. In a eudiometer 14 ml of CH_4 and 38.5 ml of O_2 was introduced. After explosion and cooling to initial temperature (23^oC) the volume of eudiometer was found to be 25 ml. Calculate the vapour pressure (in mm of Hg) of aqueous solution containing a non-volatile solute (A) at 23^oC. Mole fraction of (A) is 0.1. The pressure of eudiometer is 1 atm and is constant.

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If $f(\theta)=\sin\theta+\cos\theta+\tan\theta+\cot\theta-\sec\theta-\operatorname{cosec}\theta;\ \theta\in\left(0,\frac{\pi}{2}\right)$, then

(A) $f(\theta)>1$ (B) $f(\theta)\le\sqrt{2}$

(C) $f(\theta)\le 2$ (D) $f(\theta)<1$

38. For a certain complex number c, the polynomial $P(x)=\left(x^2-2x+2\right)\left(x^2-cx+4\right)\left(x^2-4x+8\right)$ has exactly 4 distinct roots. The value of $|c|$ is.

(A) 2 (B) $\sqrt{6}$

(C) $2\sqrt{2}$ (D) $\sqrt{10}$

39. Let α,β,γ are roots of $x^3+px+r=0$ and $\frac{\alpha}{2},\beta,2\gamma$ are the roots of equation $x^3+qx+r=0$, then

(A) $\left(\frac{q}{7}\right)^3+\left(\frac{r}{6}\right)^2=0$ (B) $p=2q$

(C) $(pq)=5r^{\frac{2}{3}}$ (D) None of these

Space for Rough work

40. If area of triangle whose vertices are $(a,a^2),(b,b^2),(c,c^2)$ is 1 and area of another triangle whose vertices are $(\alpha^2,\alpha),(\beta^2,\beta)$ and (γ^2,γ) is 2, then the absolute value of $\begin{vmatrix}(a+\alpha)^2 & (b+\alpha)^2 & (c+\alpha)^2\\ (a+\beta)^2 & (b+\beta)^2 & (c+\beta)^2\\ (a+\gamma)^2 & (b+\gamma)^2 & (c+\gamma)^2\end{vmatrix}$ is equal to:

(A) 2 (B) 4
(C) 8 (D) 16

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. Which of the following statement(s) is/are correct?

(A) If the discriminant of the quadratic trinomial $P(x)=ax^2+bx+c(a,b,c\in R)$ vanishes then the range of $P(x)$ is $[0,\infty)$

(B) If $\sin^2 x+\sec^2 y=1$, then number of ordered pairs (x,y) of real number where $x,y,\in[0,2\pi]$ is equal to 9

(C) If triangle ABC, let $a,b\,and\,c$ denote the lengths of the sides opposite to vertices $A,B\,and\,C$ respectively. If $(a+c-b)(a-c+b)=bc$, then angle A is equal to $\frac{\pi}{3}$.

(D) If $a,b>0$ then minimum value of $\left(\frac{8a}{b}+\frac{b}{2a}\right)$ is equal to 4.

Space for Rough work

42. A sequence of number is defined recursively by $a_1 = 1, a_2 = \frac{3}{7}$ and $a_n = \frac{a_{n-2} \cdot a_{n-1}}{2a_{n-2} - a_n - 1}$ for all $n \geq 3$. Then

(A) $\left|\frac{1}{a_{2019}} - \frac{1}{a_{2018}}\right| = \frac{4}{3}$

(B) $a_{2019} = \frac{3}{8075}$

(C) $\left|\frac{1}{a_{2019}} - \frac{1}{a_{2018}}\right| = \frac{3}{4}$

(D) $a_{2019} = \frac{4}{8075}$

43. If $f : \mathbb{R} \to \mathbb{R}$ is defined as $f(x) = \log_2\left(\frac{3\sin x - 4\cos x + 15}{10}\right)$, then:

(A) $f(x)$ is not injective

(B) $f(x)$ is surjective

(C) If range of $f(x)$ also satisfies the inequality $-t^2 - 2at + a < 0$, then $a \in (-1, 0)$

(D) Minimum value of $g(x) = \frac{1}{\sin^4 x + \cos^4 x}$ defined on real numbers is equal to maximum value of $f(x)$

44. Let x, y, z be real numbers with $x \geq y \geq z \geq \frac{\pi}{12}$, such that $x + y + z = \frac{\pi}{2}$ and let $P = \cos x . \sin y . \cos z$, then

(A) Minimum value of P is $\frac{1}{8}$

(B) Maximum value of P is $\frac{2+\sqrt{3}}{8}$

(C) Minimum value of P is $\frac{1}{4}$

(D) Maximum value of P is $\frac{2+\sqrt{3}}{4}$

Space for Rough work

45. If $\left|\sum_{r=1}^{n} a_r \sin(rx)\right| \le |\sin x| \forall x \in R$ for real numbers $a_1 a_2 .....a_n$, then $\left|\sum_{r=1}^{n} r\, a_r\right|$ can be equal to

(A) $\frac{1}{\pi}$ (B) $\frac{2}{\pi}$

(C) 2 (D) 1

46. Let $P\left(2\sqrt{3}, \frac{5}{2}\right)$ is any point on the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ having centre at C. Let $Q \text{ and } R$ are feet of perpendicular from foci $S_1 \text{ and } S_2$ on the tangent at P, tangent at P intersect y-axis at $T \text{ and } N$ is foot of perpendicular from centre C on the normal at P and normal at P intersect major and minor axes at $A \text{ and } B$, then

(A) $(CA)(CT)=9$ (B) $(PN)(PB)=25$

(C) $(S_1Q)(S_2R)=16$ (D) $(S_1P)(S_2P)=\frac{90}{4}$

47. Let $PA \text{ and } PB$ be the pair of tangents drawn to hyperbola $9x^2-4y^2=36$ from point $P(2,-1)$, then

(A) Equation of tangent from point P is $x+2=0$

(B) Equation of tangent from point P is $x-2=0$

(C) Equation of tangent from point P is $5x+2y-8=0$

(D) Area of ΔPAB is $\frac{1}{4}$ sq units

Space for Rough work

48. Let A and B be two square matrices of order 3 given as $A=\begin{bmatrix} x & 0 & 0 \\ 0 & x^2 & 0 \\ 0 & 0 & 1/x^3 \end{bmatrix}$ and $B=\begin{bmatrix} 3 & 2 & 3 \\ 2 & 0 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ then:

(A) Minimum value of trace $\left(A^2B\right)$ is 4.

(B) If $CA^2=B,$ then minimum value of trace $\left(C\right)$ is 4

(C) $AB-BA$ is a singular matrix

(D) det. $\left(\text{adj}\left(A^{2015}B\left(A^T\right)^{2015}\right)\right)$ equals 25.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If $I=\int_1^3 \left(\tan^{-1}\left(\frac{x^2-5x+6}{x^3-6x^2+12x-7}\right)+\tan^{-1}\left(\frac{1}{x^2-4x+4}\right)dx\right)$, then find the value of $\left[\text{I}\right]$.

[Note: $\left[\text{k}\right]$ denotes greatest integer less than or equal of k]

50. If $f\left(y\right)=\lim_{n\to\infty}\sum_{n=1}^{n}\frac{e^{n+y}\left(e-1\right)}{\left(e^n-1\right)\left(e^{n+1}-1\right)}$ then find the value of $\int_0^1 f\left(x\right)dx$.

Space for Rough work

51. Let the line $y = 100x - 199$ be intersect the graph of a function $f(x) = x^3 - 6x^2 + ax - 2a + 17, a \in R$ at three distinct points whose abscissae are x_1, x_2 and x_3 $(x_1 < x_2 < x_3)$ such that $x_3 - x_1 = 6$. If $\int_{x_1}^{x_3} (f(x) - 98x + 198)\,dx = \lambda$, then find the value of λ.

52. Let $S_n = \sum_{k=0}^{n} (-4)^k \cdot {}^{n+k}C_{2k}$, then value of $S_{2011} + 2.S_{2010} + S_{2009}$ is _____

53. Six identical balls are randomly dropped into three boxes A, B, C which can hold any number of balls. If the probability that one of the box contain 1 and the other two contains 2 and 3 balls is expressed as $\frac{p}{q}$ in lowest rational. Find $(p+q)$:

54. Sum of solution of the equation $\cos^{-1}\frac{x^2-1}{x^2+1} + \tan^{-1}\frac{2x}{x^2-1} = \frac{2\pi}{3}$ is _____

Space for Rough work